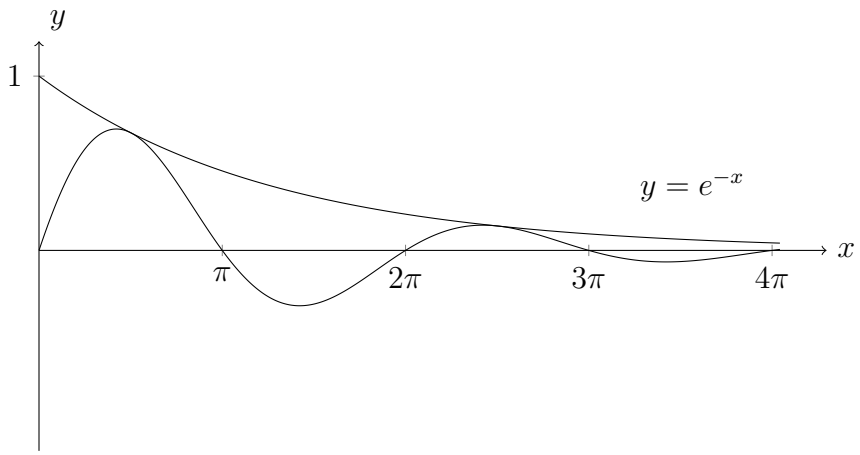


Contact points:

$$e^{-x} = e^{-x}\sin x$$

$$\sin x = 1 \implies x_n = \frac{(4n-3)\pi}{2}$$

$$\begin{aligned} A_n &= \int_{x_n}^{x_{n+1}} e^{-x}(1-\sin x)\,dx \\ &= \left[-e^{-x} + \frac{1}{2}e^{-x}(\sin x + \cos x)\right]_{x_n}^{x_{n+1}} \\ &= \frac{1}{2}\left(e^{-\frac{(4n-3)\pi}{2}} - e^{-\frac{(4n+1)\pi}{2}}\right) \\ &= \frac{1}{2}(e^{2\pi}-1)e^{-(4n+1)\pi/2} \end{aligned}$$

$$\begin{aligned} \sum_{n=1}^{\infty} A_n &= \frac{1}{2}(e^{2\pi}-1)\sum_{n=1}^{\infty} e^{-\frac{(4n+1)\pi}{2}} \\ &= \frac{1}{2}(e^{2\pi}-1)\cdot\frac{e^{-\frac{5\pi}{2}}}{1-e^{-2\pi}} \\ &= \frac{1}{2}e^{-\frac{\pi}{2}} \end{aligned}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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